1 I

1.1 (1)

The charge carried by an infinitesimal segment of the ring is $dq = qa\,d\theta$. Thus,

$$d\phi(\vec{r}) = \frac{qa\,d\theta}{4\pi\epsilon_0\sqrt{a^2+z^2}}.$$

Integrating over the entire ring, we obtain

$$\boxed{\phi(\vec{r}) = \int_0^{2\pi} d\phi(\vec{r}) = \frac{aq}{2\epsilon_0\sqrt{a^2+z^2}}}.$$

1.2 (2)

The electric field along the z-axis is

$$\boxed{\vec{E}(\vec{r}) = -\nabla\phi(\vec{r}) = \frac{qaz}{2\epsilon_0(a^2+z^2)^{3/2}}\hat{z}}.$$

At the origin,

$$\vec{E}(0,0,0) = 0.$$

Thus, the force acting on a point charge placed at the origin is

$$\boxed{\vec{F}(O) = 0}.$$

1.3 (3)

Using the first-order approximation

$$|\vec{r}-\vec{r}'|^{-3} \approx a^{-3}\left[1+\frac{3}{a}(x\cos\varphi + y\sin\varphi)\right],$$

we obtain

$$\boxed{\vec{E}(\vec{r}) = \frac{q}{4\epsilon_0 a^2}(-x,-y,2z)}.$$

In the xy-plane,

$$\vec{E}(x,y,0) = -\frac{q}{4\epsilon_0 a^2}(x,y,0).$$

For $q > 0$, the electric field points opposite to the displacement vector.

For a test charge with the opposite sign to q, the force is directed toward the origin for a small displacement along the z-axis, but away from the origin for a displacement in the xy-plane.

2 II

2.1 (4)

For $|\vec{r}| \ll a$, we can expand

$$\frac{1}{|\vec{r}-\vec{r}'|} \approx \frac{1}{a}\left[1+\frac{1}{a}(x\cos\varphi + y\sin\varphi)\right].$$

Using

$$d\vec{r}' = a(-\sin\varphi, \cos\varphi, 0)\, d\varphi,$$

the vector potential becomes

$$\vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi}\int_0^{2\pi} \frac{d\vec{r}'}{|\vec{r}-\vec{r}'|}.$$

Evaluating the integral, we obtain

$$\boxed{\vec{A}(\vec{r}) = \frac{\mu_0 I}{4a}(-y, x, 0)}.$$

2.2 (5)

The magnetic field is given by

$$\vec{B} = \nabla \times \vec{A}.$$

Thus,

$$\boxed{\vec{B} = \frac{\mu_0 I}{2a}\hat{z}}.$$

2.3 (6)

The current generated by the rotating charged ring is

$$I = \frac{2\pi a q}{2\pi/\omega} = aq\omega.$$

The angular momentum is

$$L = \mathcal{I}_C \omega.$$

Eliminating ω, we obtain

$$\boxed{I = \frac{aqL}{\mathcal{I}_C}}.$$

Since $\vec{L} = L\hat{z}$, substituting this expression into (5) gives

$$\boxed{\vec{B} = \frac{\mu_0 q}{2\mathcal{I}_C}\vec{L}}.$$

Therefore, the interaction energy between the magnetic moment and the magnetic field is

$$\boxed{V = -\vec{M}\cdot\vec{B} = -\frac{\mu_0 q}{2\mathcal{I}_C}\vec{M}\cdot\vec{L}}.$$

3 III

3.1 (7)

The magnetic field can be obtained from the vector potential:

$$\vec{B} = \nabla \times \vec{A}.$$

Using $\vec{m}_1 = (0, 0, \pi I a^2)$, we obtain

$$\begin{aligned}\vec{B}(\vec{r}) &= \frac{\mu_0}{4\pi r^5}\left[3(\vec{m}_1 \cdot \vec{r})\vec{r} - \vec{m}_1 r^2\right] \\ &= \frac{3\mu_0 I a^2}{4r^5}\left(xz, yz, z^2 - \frac{r^2}{3}\right).\end{aligned}$$

This agrees with the magnetic field of a magnetic dipole.

3.2 (8)

Along the z-axis, the magnetic field is

$$\vec{B}^{(z)} = \frac{\mu_0}{2\pi r^3}\vec{m}_1.$$

For $\vec{m}_2 = \pm\vec{m}_1$, the interaction energy is

$$\boxed{V^{(z)} = \mp\frac{\mu_0 \pi I^2 a^4}{2r^3}}.$$

In the xy-plane, since $\vec{m}_1 \cdot \vec{r} = 0$, the magnetic field becomes

$$\vec{B}^{(xy)} = -\frac{\mu_0}{4\pi r^3}\vec{m}_1.$$

Therefore,

$$\boxed{V^{(xy)} = \pm\frac{\mu_0 \pi I^2 a^4}{4r^3}}.$$

Here, the upper and lower signs correspond to $\vec{m}_2 = +\vec{m}_1$ and $\vec{m}_2 = -\vec{m}_1$, respectively.